ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech. (CSE-AI)

Semester: VI

SUBJECT CODE: BCSAI606

BLOCKCHAIN

Time: 02:00 Hrs. **Maximum Marks: 35**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. **(07 Marks)**

Q. 1. Define the following terms: **(1 × 7 = 7 Marks)**

(i) What do you mean by cipher text?

(ii) Defined crypto-currency.

(iii) Define Integrity?

(iv) What is the application of Digital Signatures?

(v) What are the Advantage and disadvantages of Blockchain?

(vi) Give two examples in which blockchain is used.

(vii) What is IDE?

SECTION – B

NOTE:Attempt all questions. **(3.5×2×2 = 14 Marks)**

Q.2.(i)(a) How does distributed consensus work in blockchain networks?

OR

(**b)** Compare symmetric and asymmetric encryption methods.

(ii)(a) Define Lifetime of Keys, Destroying Keys and Public-Key Management.

OR

(b) How does Ethereum differ from Bitcoin?

Q.3.(i)(a) Illustrate development machine specifications and IDE.

OR

(**b)** What are the various methods for generating cryptographic keys?

(ii)(a) What is traffic confidentiality? Also write its importance.

OR

(**b)** Describe the Ethereum Eco-System in brief.

SECTION – C

NOTE: Attempt all questions. **(7 × 2 = 14 Marks)**

Q.4.(a) Describe in brief about RSA.

OR

(b) Describe the role of cryptography and hashing in blockchain security.

Q.5.(a) Distinguish between Substitution Techniques and Transposition Techniques.

OR

(b) How do digital signatures ensure security in blockchain transactions? Discuss in detail.